

60-E

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING, LU

Computer Graphics Sessional

ASSIGNMENT

Bridge to the Real World

Field	Details
Assignment Title	Bridge to the Real World — Design & Build
Course	Computer Graphics Sessional
Submission Format	PDF Report + Source Code (GitHub link) + Live Viva
Viva Duration	6 minutes per student (2 min demo + 4 min Q&A)

Note

This assignment is not about grades. It is about proof — proof to yourself, that you can take a real problem, apply what you have learned, and build something that works.

You will need to say: **“I built this.”**

One rule: you must build it yourself.

Assignment Overview

Each student selects a real-world problem that can be solved using Computer Graphics techniques, then designs and implements a working prototype in Python OpenGL.

Part	Title	What You Produce
Part 1	Problem Discovery	1-page written justification of your chosen project
Part 2	Implementation	Working Python OpenGL prototype with inline comments
Part 3	Reflection Report	Personal written report answering 5 deep questions
Part 4	Viva	Live + Q&A

Choose Your Career Track

Track	If You Are Heading Toward	Suggested Project Domain
Track A	Software Job (local or abroad)	UI renderer, 2D map viewer, data visualisation engine
Track B	Higher Study — MS / PhD abroad	Algorithm comparison tool, research visualiser
Track C	Game / Graphics Industry or Freelancing	Mini 2D game, animation, interactive scene

Note: You may propose a project outside these examples — but you must justify it in Part 1.

Part 1 — Problem Discovery

Submit a one-page written document answering:

1. Which career track have you chosen?
2. What is your project idea?
3. Which Computer Graphics concepts will your project use?
4. Find two real-world systems using similar techniques.

Part 2 — Implementation

Build a working prototype using Python OpenGL (PyOpenGL + GLUT).

Required Techniques (Use Any 4 at least in your project)

Technique	Real-World Connection
Line & Shape Drawing	Google Maps, vector graphics
2D Transformations	Game engines and animation
Color Fill	Photoshop paint bucket
Line Clipping	GPU rendering pipeline
Bezier Curves	Font design, automotive modelling

Mandatory Comment Style

কী করছে: ...

কেন লাগছে: ...

real world-এ এটা কোথায় দেখা যায়: ...

Important — During viva/presentation, you must modify your code live.

Part 3 — Reflection Report

The Five Questions

Q1. Where did you get most stuck?

Explain the bug/problem and how you solved it.

Q2. What did this project teach you that classroom theory could not?

Discuss practical lessons learned.

Q3. How does this connect to your career track?

Relate your work to jobs, research, or game development.

Q4. If you had one more week, what would you add?

Describe future improvements.

Q5. Would you recommend this type of projects in CG lab?

Be honest with justifications.

Part 4 — Viva / Presentation

Session	Duration	What Happens
Demo	2 min	Live demonstration
Q&A	4 min	Teacher's Questions

Sample Viva Questions

- “Delete this line of code. What happens?”
- “What is the weakest part of your project?”
- “Explain your project to a non-technical person.”
- “Walk me through one function.”

Bonus Rule

“While implementing this, I discovered something never covered in the past...”

If you explain that discovery clearly during Viva, you earn bonus marks.

Academic Integrity

You may use:

- Documentation
- Tutorials
- AI tools for learning/debugging

However:

- Your submitted code must be your own
- Your reflection must be personal
- You must defend your code live during viva